

Divergence of a vector Point Function

The divergence of a differentiable vector point function $\vec{V}$ is denoted by $\text{div } \vec{V}$ and is defined as

$$\text{div } \vec{V} = \nabla \cdot \vec{V} = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot \vec{V} = \hat{i} \cdot \frac{\partial \vec{V}}{\partial x} + \hat{j} \cdot \frac{\partial \vec{V}}{\partial y} + \hat{k} \cdot \frac{\partial \vec{V}}{\partial z}$$

The divergence of a vector point function is a scalar point fn.

* If $\nabla \cdot \vec{V} = 0$, the vector field $\vec{V}$ is called SOLENOIDAL.

Curl of a vector Point function

The curl (or rotation) of a differentiable vector point function $\vec{V}$ is denoted by $\text{curl } \vec{V}$ and is defined as

$$\begin{aligned} \text{curl } \vec{V} &= \nabla \times \vec{V} = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \times \vec{V} \\ &= \hat{i} \times \frac{\partial \vec{V}}{\partial x} + \hat{j} \times \frac{\partial \vec{V}}{\partial y} + \hat{k} \times \frac{\partial \vec{V}}{\partial z} \end{aligned}$$

The curl of a vector point fn is a vector point fn.

* If $\text{curl } \vec{V} = 0$, then $\vec{V}$ is said to be irrotational vector, otherwise rotational. i.e., $\nabla \times \vec{V} = 0$

If $\vec{V}$ is irrotational there exists a scalar point fn ϕ such that $\vec{V} = \nabla \phi$ and ϕ is called the scalar Potential of $\vec{V}$.

* The formula for finding scalar potential ϕ for an irrotational field $\vec{V}$ is given by

$d\phi = V_1 dx + V_2 dy + V_3 dz$ which is directly integrable

$$\begin{aligned} \text{If } \vec{V} &= V_1 \hat{i} + V_2 \hat{j} + V_3 \hat{k} \text{ then } \text{curl } \vec{V} = \nabla \times \vec{V} = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \times (V_1 \hat{i} + V_2 \hat{j} + V_3 \hat{k}) \\ &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ V_1 & V_2 & V_3 \end{vmatrix} = \hat{i} \left(\frac{\partial V_3}{\partial y} - \frac{\partial V_2}{\partial z} \right) + \hat{j} \left(\frac{\partial V_1}{\partial z} - \frac{\partial V_3}{\partial x} \right) \\ &\quad + \hat{k} \left(\frac{\partial V_2}{\partial x} - \frac{\partial V_1}{\partial y} \right) \end{aligned}$$

Q. If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, show that

i. $\text{div } \vec{r} = 3$ (ii) $\text{curl } \vec{r} = 0$

Solⁿ. (i) $\text{div } \vec{r} = \nabla \cdot \vec{r} = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot (x\hat{i} + y\hat{j} + z\hat{k})$
 $= 1 + 1 + 1$
 $\begin{cases} \hat{i} \cdot \hat{i} = 1 \\ \hat{j} \cdot \hat{j} = 1 \\ \hat{k} \cdot \hat{k} = 1 \end{cases}$

(ii) $\text{curl } \vec{r} = \nabla \times \vec{r} = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \times (x\hat{i} + y\hat{j} + z\hat{k})$

$= \hat{k} \frac{\partial(y)}{\partial x} - \hat{j} \frac{\partial(z)}{\partial x} - \hat{k} \frac{\partial(x)}{\partial y} + \hat{i} \frac{\partial(z)}{\partial y}$

$+ \hat{j} \frac{\partial(x)}{\partial z} - \hat{i} \frac{\partial(y)}{\partial z} = \hat{i}(0) + \hat{j}(0) + \hat{k}(0)$

$= 0$

$\begin{cases} \hat{i} \times \hat{i} = 0 \\ \hat{j} \times \hat{j} = 0 \\ \hat{k} \times \hat{k} = 0 \\ \hat{i} \times \hat{j} = \hat{k} \end{cases}$



Q. Find the divergence and curl of a vector

$\vec{v} = (xyz)\hat{i} + (3x^2y)\hat{j} + (xz^2 - y^2z)\hat{k}$ at point $(2, -1, 1)$

Solⁿ. $\text{div } \vec{v} = \nabla \cdot \vec{v} = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot ((xyz)\hat{i} + (3x^2y)\hat{j} + (xz^2 - y^2z)\hat{k})$

$= \frac{\partial}{\partial x}(xyz) + \frac{\partial}{\partial y}(3x^2y) + \frac{\partial}{\partial z}(xz^2 - y^2z)$

$= yz + 3x^2 + 2xz - y^2$

$= -1 + 12 + 4 - 1 = 14$ at $(2, -1, 1)$

$\text{curl } \vec{v} = \nabla \times \vec{v} = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \times ((xyz)\hat{i} + (3x^2y)\hat{j} + (xz^2 - y^2z)\hat{k})$

$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xyz & 3x^2y & xz^2 - y^2z \end{vmatrix} = \hat{i}(-2yz - 0) + \hat{j}(xy - z^2) + \hat{k}(6xy - xz)$
 $= 2\hat{i} - 3\hat{j} - 14\hat{k}$ at $(2, -1, 1)$

Q. Find $\text{div } \vec{F}$ and $\text{curl } \vec{F}$, where $\vec{F} = \text{grad } (x^3 + y^3 + z^3 - 3xyz)$

Solⁿ: $\phi = x^3 + y^3 + z^3 - 3xyz$, then

$$\vec{F} = \text{grad } \phi = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot (x^3 + y^3 + z^3 - 3xyz)$$

$$\vec{F} = (3x^2 - 3yz)\hat{i} + (3y^2 - 3xz)\hat{j} + (3z^2 - 3xy)\hat{k}$$

$$\begin{aligned} \text{Now, } \text{div } \vec{F} = \nabla \cdot \vec{F} &= \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot \left\{ (3x^2 - 3yz)\hat{i} \right. \\ &\quad \left. + (3y^2 - 3xz)\hat{j} + (3z^2 - 3xy)\hat{k} \right\} \\ &= \frac{\partial}{\partial x} (3x^2 - 3yz) + \frac{\partial}{\partial y} (3y^2 - 3xz) + \frac{\partial}{\partial z} (3z^2 - 3xy) \end{aligned}$$

$$= 6x + 6y + 6z$$

$$= 6(x + y + z)$$

$$\text{curl } \vec{F} = \nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ (3x^2 - 3yz) & (3y^2 - 3xz) & (3z^2 - 3xy) \end{vmatrix}$$

$$= \hat{i}(-3x + 3x) + \hat{j}(-3y + 3y) + \hat{k}(-3z + 3z) = 0$$

④ Properties of Divergence and Curl

- ① For a const. vector $\vec{a}$,
 $\text{div } \vec{a} = 0$ and $\text{curl } \vec{a} = 0$
- ② $\text{div}(\vec{A} + \vec{B}) = \text{div } \vec{A} + \text{div } \vec{B}$ or $\nabla \cdot (\vec{A} + \vec{B}) = \nabla \cdot \vec{A} + \nabla \cdot \vec{B}$
- ③ $\text{curl}(\vec{A} + \vec{B}) = \text{curl } \vec{A} + \text{curl } \vec{B}$ or $\nabla \times (\vec{A} + \vec{B}) = \nabla \times \vec{A} + \nabla \times \vec{B}$
- ④ If $\vec{A}$ is a vector fn and ϕ is a scalar^{fn}, then
 $\text{div}(\phi \vec{A}) = \phi \text{div } \vec{A} + (\text{grad } \phi) \cdot \vec{A}$
or $\nabla \cdot (\phi \vec{A}) = \phi (\nabla \cdot \vec{A}) + (\nabla \phi) \cdot \vec{A}$
- ⑤ If $\text{curl}(\phi \vec{A}) = (\text{grad } \phi) \times \vec{A} + \phi \text{curl } \vec{A}$
or $\nabla \times (\phi \vec{A}) = (\nabla \phi) \times \vec{A} + \phi (\nabla \times \vec{A})$.
- ⑥ $\nabla(\vec{A} \cdot \vec{B}) = (\vec{A} \cdot \nabla) \vec{B} + (\vec{B} \cdot \nabla) \vec{A} + \vec{A} \times (\nabla \times \vec{B}) + \vec{B} \times (\nabla \times \vec{A})$.
- ⑦ $\nabla \cdot (\vec{A} \times \vec{B}) = \vec{B} \cdot (\nabla \times \vec{A}) - \vec{A} \cdot (\nabla \times \vec{B})$
- ⑧ $\nabla \times (\vec{A} \times \vec{B}) = (\nabla \cdot \vec{B}) \vec{A} - (\nabla \cdot \vec{A}) \vec{B} + (\vec{B} \cdot \nabla) \vec{A} - (\vec{A} \cdot \nabla) \vec{B}$

Repeated Operations by ∇ .

- ① $\text{Div}(\text{grad } \phi) = \nabla^2 \phi$, where $\nabla^2 \phi = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \phi$.
- ② $\text{curl}(\text{grad } \phi) = \nabla \times \nabla \phi = \vec{0}$
- ③ $\text{Div}(\text{curl } \vec{V}) = \nabla \cdot (\nabla \times \vec{V}) = \vec{0}$
- ④ $\text{curl}(\text{curl } \vec{V}) = \text{grad div } \vec{V} - \nabla^2 \vec{V}$.

Q. A vector field is given by $\vec{A} = (x^2 + xy^2)\hat{i} + (y^2 + x^2y)\hat{j}$.

Show that the field is irrotational

Solⁿ Field A is irrotational if $\text{curl } \vec{A} = 0$

$$\text{Now, } \text{curl } \vec{A} = \nabla \times \vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 + xy^2 & y^2 + x^2y & 0 \end{vmatrix}$$

$$= \hat{i}(0-0) + \hat{j}(0-0) + \hat{k}(2xy-2xy) = 0$$

$\therefore$ Field $\vec{A}$ is irrotational

Q. A vector field $\vec{F}$ is given by $\vec{F} = (x^2 - y^2 + x)\hat{i} + (2xy + y)\hat{j}$

Show that the field is irrotational and find its scalar potential

Solⁿ $\vec{F} = (x^2 - y^2 + x)\hat{i} + (2xy + y)\hat{j}$

For $\vec{F}$ to be irrotational $\nabla \times \vec{F} = 0$

$$\text{Here } \nabla \times \vec{F} = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} \right) \times \left[(x^2 - y^2 + x)\hat{i} + (2xy + y)\hat{j} \right]$$

$$= -2y\hat{k} - \hat{k}(-2y) = -2y\hat{k} + 2y\hat{k} = 0$$

$\therefore \vec{F}$ is irrotational. To obtain scalar potential ϕ , say, we have $\vec{F} = \nabla \phi$

$$\Rightarrow \frac{\partial \phi}{\partial x} = x^2 - y^2 + x \quad \text{--- (1)} \quad \frac{\partial \phi}{\partial y} = -2xy - y \quad \text{--- (2)}$$

Integrating (1) w.r.t. x , we get $\phi = \frac{x^3}{3} - xy^2 + \frac{x^2}{2} + C(y)$

$$\therefore \frac{\partial \phi}{\partial y} = -2xy + \frac{dC}{dy}$$

But $\frac{\partial \phi}{\partial y} = -2xy - y$ Hence $\frac{dC}{dy} = -y \Rightarrow C = -\frac{y^2}{2}$

$\therefore \phi = \frac{x^3}{3} - xy^2 + \frac{x^2}{2} - \frac{y^2}{2}$ is the req^d scalar potential

Q. Show that the vector field $\vec{F} = \frac{\vec{r}}{r^3}$ is irrotational as well as solenoidal. Also find the scalar potential

$\xrightarrow{\text{Sol}^n}$ $\vec{F} = \frac{\vec{r}}{r^3} = \frac{x\hat{i} + y\hat{j} + z\hat{k}}{r^3}$, hence

$$\begin{aligned} \text{div } \vec{F} = \nabla \cdot \vec{F} &= \sum \frac{\partial}{\partial x} \left(\frac{x}{r^3} \right) = \sum \left(\frac{1}{r^3} - \frac{3x}{r^4} \cdot \frac{x}{r} \right) \\ &= \frac{3}{r^3} - \frac{3}{r^5} (x^2 + y^2 + z^2) = 0 \end{aligned}$$

$\therefore \vec{F}$ is a solenoidal vector.

Q. If $\vec{A} = x^2 y z \hat{i} + x y z^2 \hat{j} + y^2 z \hat{k}$ determine $\text{curl } \vec{A}$.

$\xrightarrow{\text{Sol}^n}$ we know that $\nabla \times \nabla \times \vec{A} = \nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A}$.

Here $\vec{A} = x^2 y z \hat{i} + x y z^2 \hat{j} + y^2 z \hat{k}$

Now, $\nabla \cdot \vec{A} = \frac{\partial}{\partial x} (x^2 y z) + \frac{\partial}{\partial y} (x y z^2) + \frac{\partial}{\partial z} (y^2 z) = 2x y z + x z^2 + y^2$

$$\begin{aligned} \therefore \nabla(\nabla \cdot \vec{A}) &= \hat{i} \frac{\partial}{\partial x} (2x y z + x z^2 + y^2) + \hat{j} \frac{\partial}{\partial y} (2x y z + x z^2 + y^2) \\ &\quad + \hat{k} \frac{\partial}{\partial z} (2x y z + x z^2 + y^2) \end{aligned}$$

$$= \hat{i} (2y z + z^2) + \hat{j} (2x z + 2y) + \hat{k} (2x y + 2x z)$$

and $\nabla^2 \vec{A} = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) (x^2 y z \hat{i} + x y z^2 \hat{j} + y^2 z \hat{k})$

$$= (2y z \hat{i} + 0 \hat{j} + 0 \hat{k}) + (0 \hat{i} + 0 \hat{j} + 2z \hat{k}) + (0 \hat{i} + 2xy \hat{j} + 0 \hat{k})$$

$$= 2y z \hat{i} + 2xy \hat{j} + 2z \hat{k}$$

$$\begin{aligned} \therefore \nabla \times (\nabla \times \vec{A}) &= \hat{i} (2y z + z^2) + \hat{j} (2x z + 2y) + \hat{k} (2x y + 2x z) \\ &\quad - (2y z \hat{i} + 2xy \hat{j} + 2z \hat{k}) \\ &= \hat{i} z^2 + \hat{j} (2x z + 2y - 2xy) + \hat{k} (2xy + 2xz - 2z) \end{aligned}$$

Directional Derivative

$\frac{\partial \phi}{\partial s}$ represents the rate of change of ϕ in any direction and is called directional derivative. The directional derivative in the direction of vector $\vec{a}$ is $\nabla \phi \cdot \hat{a}$. The directional derivatives of ϕ is maximum along the normal to the level surface of ϕ and is equal to $|\nabla \phi| = \frac{\partial \phi}{\partial n}$.

Q. Compute the directional derivative of $x^2 + y^2 + 4xz$ at $(1, -2, 2)$ in the direction of the vector $2\hat{i} - 2\hat{j} - \hat{k}$

Solⁿ. The given scalar fn is $f = x^2 + y^2 + 4xz$

$$\text{Now, } \nabla f = \hat{i} \frac{\partial f}{\partial x} + \hat{j} \frac{\partial f}{\partial y} + \hat{k} \frac{\partial f}{\partial z} = (2x + 4z)\hat{i} + (2y)\hat{j} + 4x\hat{k}$$

$$= 10\hat{i} - 4\hat{j} + 4\hat{k} \text{ at } (1, -2, 2)$$

The directional derivative of f at $(1, -2, 2)$ along the vector $(2\hat{i} - 2\hat{j} - \hat{k})$

$$= (10\hat{i} - 4\hat{j} + 4\hat{k}) \cdot \frac{2\hat{i} - 2\hat{j} - \hat{k}}{\sqrt{4+4+1}}$$

$$= \frac{1}{3} (20 + 8 - 4) = \frac{24}{3} = \underline{8}$$

Q. If $f(x, y) = x^2 - xy - y + y^2$ show that all the points where the directional derivative in the direction of $\vec{b} = \frac{\hat{i} + \sqrt{3}\hat{j}}{2}$ is zero, lie on a line. Find the equation of the line.

Solⁿ Given that $f = x^2 - xy - y + y^2$, then $\nabla f = \hat{i} \frac{\partial f}{\partial x} + \hat{j} \frac{\partial f}{\partial y} + \hat{k} \frac{\partial f}{\partial z}$

$$= \hat{i}(2x - y) + \hat{j}(-x - 1 + 2y)$$

Here $\hat{b} = (i + \sqrt{3}j)/2$

$\therefore$ Directional derivative of f along $\hat{b}$

$$= \nabla f \cdot \left(\frac{i + \sqrt{3}j}{2} \right) = \frac{1}{2} [(2x - y) + \sqrt{3}(-x - 1 + 2y)]$$

$\therefore$ this directional derivative is given to be zero, we have

$$2x - y = 0 \quad \text{and} \quad -x - 1 + 2y = 0 \Rightarrow x = \frac{1}{3}, y = \frac{2}{3}$$

Since the line is along a vector $\hat{b}$ its slope is $\sqrt{3}$.

$\therefore$ Eqn of the line is $y - \frac{2}{3} = \sqrt{3}(x - \frac{1}{3})$

Q. Find the directional derivative of the scalar f
 $f(x, y, z) = x^2 + xy + z^2$ at the point $A(1, -1, -1)$ in the direction of the line AB where B has the coordinates $(3, 2, 1)$

Solⁿ $f = x^2 + xy + z^2$, $\therefore \nabla f = \hat{i} \frac{\partial f}{\partial x} + \hat{j} \frac{\partial f}{\partial y} + \hat{k} \frac{\partial f}{\partial z}$

$$= (2x + y)\hat{i} + x\hat{j} + 2z\hat{k}$$

$\therefore (\nabla f)_A = [\nabla f]_{(1, -1, -1)} = (2 - 1)\hat{i} + 1\hat{j} + 2(-1)\hat{k} = \hat{i} + \hat{j} - 2\hat{k}$

Also $\vec{AB} = (3 - 1)\hat{i} + (2 + 1)\hat{j} + (1 + 1)\hat{k} = 2\hat{i} + 3\hat{j} + 2\hat{k}$

$\therefore$ The DD of f at A along $\vec{AB}$

$$= [\nabla f]_{(1, -1, -1)} \cdot \hat{AB} = (\hat{i} + \hat{j} - 2\hat{k}) \cdot \frac{(2\hat{i} + 3\hat{j} + 2\hat{k})}{\sqrt{2^2 + 3^2 + 2^2}}$$

$$= \frac{1}{\sqrt{17}} (2 + 3 - 4) = \frac{1}{\sqrt{17}}$$